



Mesh Current Method (Mesh Analysis)

A mesh is a loop which does not contain any other loops.

For mesh analysis, we take the mesh currents as the circuit variables. Mesh current is a fictitious quantity and is not necessarily the branch current.

A) For a circuit **without any current sources**, take these steps:

1. Identify the meshes and assign a current to each mesh.
2. Apply KVL around each mesh. Use Ohm's law to express all voltages in terms of the mesh currents.
3. Solve the KVL equations to find the mesh currents.

Ex. Find V_o .

KVL in mesh 1:

$$2i_1 + 8(i_1 - i_2) + 6(i_1 - i_3) = 0$$

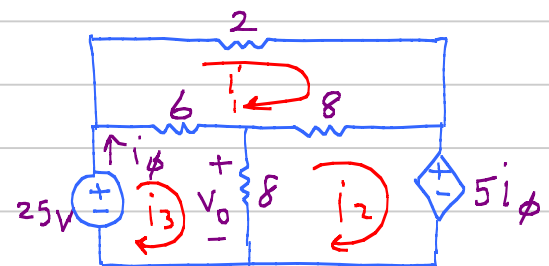
KVL in mesh 2:

$$5i_3 + 8(i_2 - i_3) + 8(i_2 - i_1) = 0$$

KVL in mesh 3:

$$-25 + 6(i_3 - i_1) + 8(i_3 - i_2) = 0$$

$$\Rightarrow i_2 = 2, \quad i_3 = 4, \quad V_o = 8(i_3 - i_2) = \boxed{16V}$$



B) when a **current source exists in the circuit**, there are two possibilities:

case 1: If the current source exists only in one mesh, set the mesh current equal to the current of the source.

Ex. Find v_ϕ using mesh analysis.

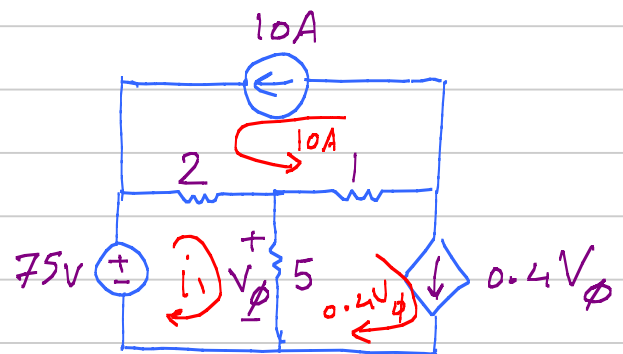
There are two current sources in the circuit which are the same as the mesh currents.

$$v_\phi = 5(i_1 - 0.4v_\phi) \Rightarrow v_\phi = \frac{5}{3} i_1$$

KVL in mesh 1:

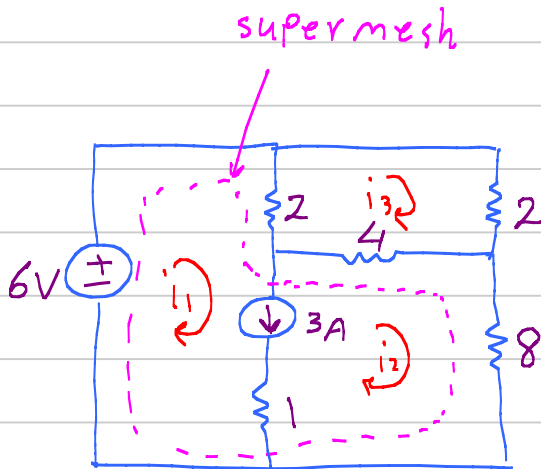
$$2(i_1 + 10) + \frac{5}{3} i_1 - 75 = 0$$

$$i_1 = 15 \text{ A} \quad , \quad v_\phi = \frac{5}{3} (15) = \underline{25 \text{ V}}$$



case 2: If the current source is shared by two meshes, create a super mesh by mentally removing the source and the elements connected in series with it. Then express the voltages around the super mesh in terms of the original mesh currents. Add a constraint equation which relates the currents of the combined meshes to the current of the source.

Ex. Find the mesh currents.



KVL in mesh 3:

$$2i_3 + 4(i_3 - i_2) + 2(i_3 - i_1) = 0$$

KVL in S.M:

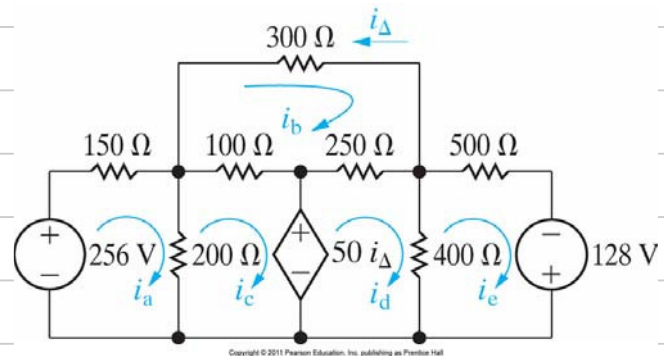
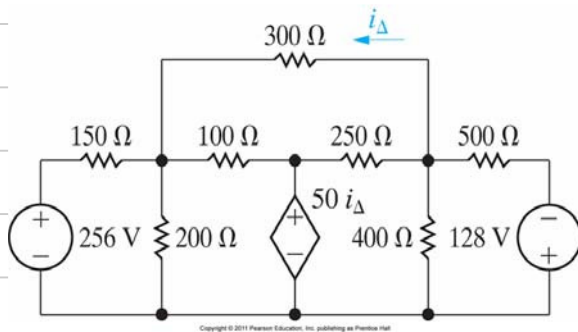
$$8i_2 - 6 + 2(i_1 - i_3) + 4(i_2 - i_3) = 0$$

constraint equation:

$$i_1 - i_2 = 3$$

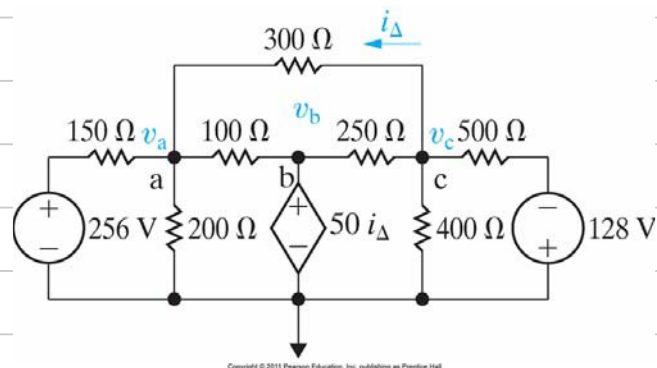
$$i_1 = 3.47 \text{ A}, \quad i_2 = 0.47 \text{ A}, \quad i_3 = 1.105 \text{ A}$$

Which method is preferred?



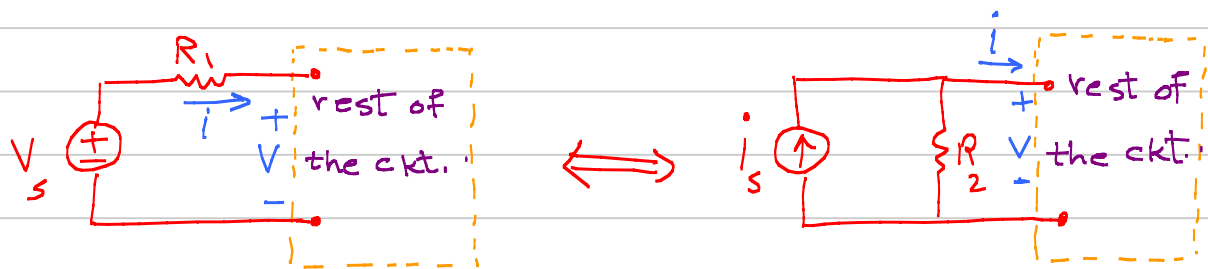
In this example, using mesh analysis, there are 5 unknown currents.

Using nodal analysis, there are only 2 unknown voltages.



Source Transformation

Source transformation is a simplification technique. It allows a voltage source in series with a resistor to be replaced by a current source in parallel with the same resistor, or vice versa.



$$-V_s + R_1 i + V = 0$$

$$V = V_s - R_1 i \quad (1)$$

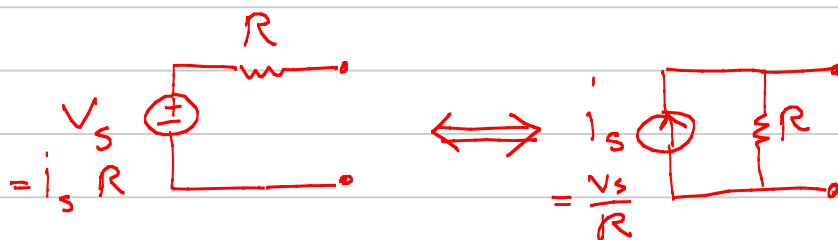
$$V = R_2 (i_s - i)$$

$$V = R_2 i_s - R_2 i \quad (2)$$

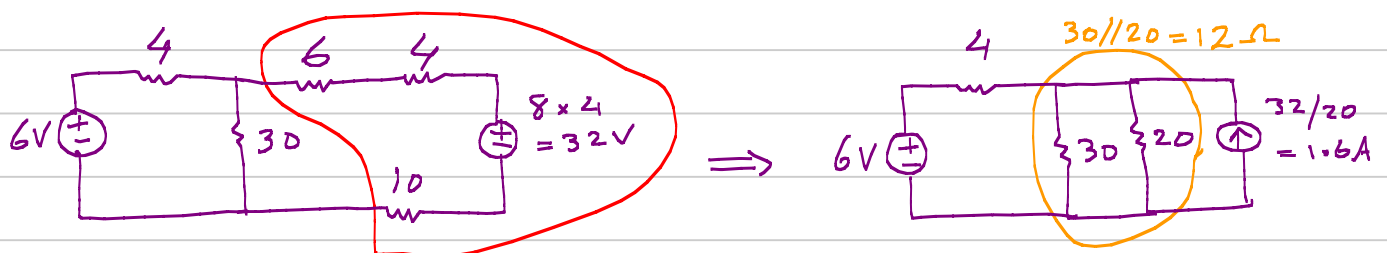
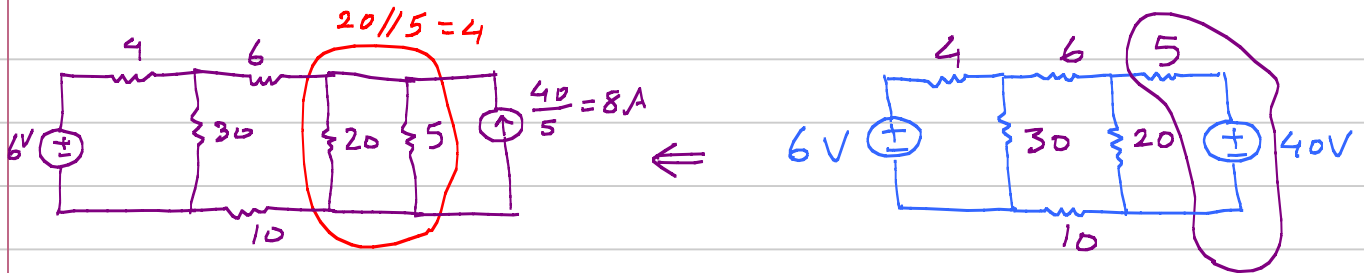
The above circuits are equivalent if they have the same $v-i$ relationship across the terminals.

$$(1) = (2) \Rightarrow V_s - R_1 i = R_2 i_s - R_2 i$$

$$\Rightarrow \boxed{R_1 = R_2 = R}, \quad \boxed{i_s = \frac{V_s}{R}}$$



Ex. Use source transformation to find the power in the 6v source.

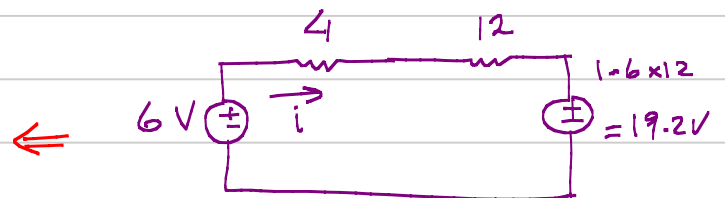


KVL:

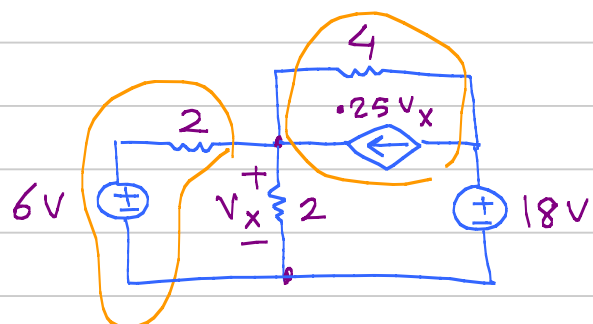
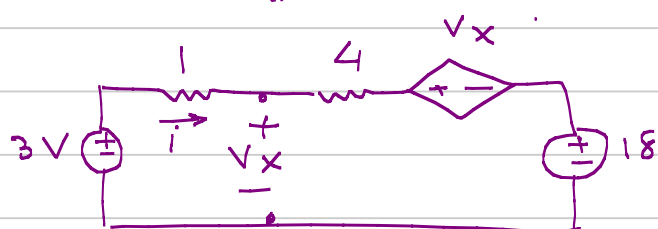
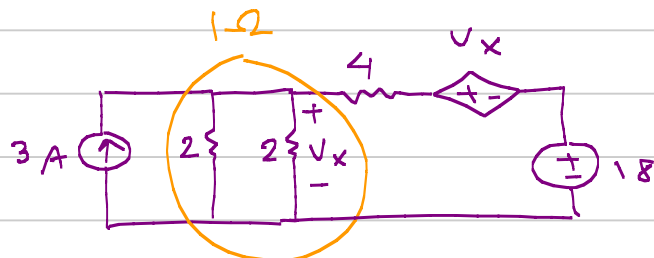
$$-6 + 16i + 19.2 = 0$$

$$i = -0.825A$$

$$P = -6i = \underline{4.95W}$$



Ex. Find v_x using source transformation.



$$-3 + 5i + v_x + 18 = 0$$

$$-3 + i + v_x = 0$$

$$\Rightarrow \underline{v_x = 7.5V}$$